

Introduction to Contract Theory and Auctions
Lecture 4
Lotteries with Monetary Prizes & Applications

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- Our discussion begin where we left last time: a DM satisfies the VNM assumptions where the space of prizes is now a set $Z \subseteq \mathbb{R}$ and $a \in Z$ is interpreted as receiving \$ a .
- In contrast to the last lecture in which we assumed the set Z was finite, in this lecture we will assume that it is an infinite set.
- Due to technical reasons we will focus on a restricted set of possible lotteries:
 - For some examples, we will use the set of lotteries which are **finitely supported** on Z , i.e. if p is a lottery, then $\text{supp}(p) \subseteq \mathbb{R}$ is a finite set of points.
 - However, for most of the lecture we will generalize our analysis to the set of lotteries consists of lotteries which are supported on a compact¹ subset of $\mathbb{R}$.
 - For many of the proofs, we will make the extra assumption that if p is a lottery, it has a well defined probability density function, i.e. p is a pdf.

¹A compact subset of $\mathbb{R}$ is closed and bounded.

- Let Z be an **infinite** set of prizes, $Z = [w, b] \subseteq \mathbb{R}$, $w < b$.
- Let $\mathfrak{F}$ be the set of CDFs on Z .
- Recall the following definition.

Definition 4.1

For any $F \in \mathfrak{F}$, the following conditions hold:

- ① $\lim_{x \rightarrow w} F(x) = 0$ and $\lim_{x \rightarrow b} F(x) = 1$.
 - ② $F(x)$ is a nondecreasing function of x .
 - ③ $F(x)$ is right-continuous; that is, for every number x_0 , $\lim_{x \downarrow x_0} F(x) = F(x_0)$.
- The set $\mathfrak{F}$ will play the role of the set of lotteries.

- Suppose $\psi : Z \rightarrow \mathbb{R}$ is a bounded function. For a $F \in \mathfrak{F}$, consider the following integral:

$$\int_Z \psi(z) dF(z), \quad (1)$$

the previous integral is in the sense of *Lebesgue–Stieltjes* and should be interpreted as the **expectation of ψ with respect to F** .

- The *Lebesgue–Stieltjes* integral generalizes the integral you are probably most familiar with, the *Riemann Integral*.
- Using the integral in (1) allows us to integrate over a larger class of functions, but most importantly, consolidates under the same concepts the “discrete” and “continuous” cases.
 - If p is a lottery with finite support in Z , i.e., it attaches positive probability $p(z) > 0$ to the finite collection of prizes $z_k \in Z$, $k = 1, \dots, K$, then $F(x) := \sum_{z_k \in Z: z_k \leq x} p(z_k)$ and the integral in (1) simplifies to

$$\int_Z \psi(z) dF(z) = \sum_{z \in Z} \psi(z) p(z)$$

- Given $F \in \mathfrak{F}$, we can define its corresponding “lottery” as follows: for a subset of prizes $A \subseteq Z$, the probability that lottery p attaches to the prize z being in A is defined as:

$$p(A) := \Pr(z \in A) = \int_Z \mathbf{1}_A(z) dF(z), \quad (2)$$

where $\mathbf{1}_A$ denotes the indicator function of the set A .

- Thus, the lottery p is a function defined over subsets of Z that attaches a probability to each subset.
- Not all subsets of Z can be assigned a probability; only a certain subclass of subsets can be assigned a “meaningful” number that can be interpreted as a probability. I am not being rigorous here on purpose, and I will brush aside this technical point to not complicate more the discussion.
- Note that in definition 4.1 it is not required for the cdf F to be fully continuous, just right continuous.

- However, in the case that the cdf F is not only right-continuous but satisfies a strong notion of continuity (absolute continuity), then we have the following definition:

Definition 4.2

The **probability density function** or pdf, $f(x)$ of the cdf $F \in \mathfrak{F}$ is the function that satisfies

$$F(x) = \int_w^x f(t)dt \quad \text{for all } x$$

- Thus, if $F \in \mathfrak{F}$ has a pdf f , then the integral in (1) can be simplified as

$$\int_w^b \psi(z)f(z)dz, \tag{3}$$

where the integral in (3) is the standard Riemann integral that you know from Calculus I.

We adjust the vNM axioms slightly to fit this framework. Consider a binary relation $\succsim$ over $\mathfrak{F}$.

Definition 4.3 (Completeness and Transitivity (A) + Monotonicity for Money (MM))

$\succsim$ is complete and transitive, i.e. a preference relation and it satisfies monotonicity for money, that is $x > y$ implies $[x] \succsim [y]$.

Definition 4.4 (Independence (I))

$p \succ q$, and $\alpha \in (0, 1)$ and implies that $\alpha p \oplus (1 - \alpha)r \succ \alpha q \oplus (1 - \alpha)r$ for any $r \in \mathfrak{F}$.

We will of course need some form of continuity axiom. The form of continuity we impose is stronger than the one we saw before.

Definition 4.5 (Continuity (C))

For every $F \in \mathfrak{F}$, the sets $\{G : G \succsim F\}$ and $\{G : F \succsim G\}$ are closed under the topology induced by the metric

$$d(F, G) = \int_{\mathbb{R}} |F(x) - G(x)| dx$$

- (C) implies that we are endowing $\mathfrak{F}$ with the weak-topology.
- This is the topology of convergence in distribution, e.g. if $F_n \rightarrow F$ at every continuity point of F and $F_n \succsim G$ for all n , then $F \succsim G$.

Example 4.6

- Consider the sequence of sure lotteries $[1 + 1/n]$, $n \in \mathbb{N}$. As you can easily see, the cdfs converge to the cdf of the lottery $[1]$.
- Consider the sequence of lotteries p_n which gives prizes $\{0, 1/n, 2/n, \dots, (n-1)/n, 1\}$, each with probability $1/(n+1)$. This sequence converges in distribution to the uniform distribution.
- Consider now the sequence of lotteries whose cdf's are the normal cdf's with mean 4 and variance $1/n$. This sequence converges in distribution to the degenerate distribution at value 4.

Theorem 4.7

The binary relation $\succsim$ satisfies A, MM, I and C if and only if there is a continuous, strictly increasing $u : Z \rightarrow \mathbb{R}$, such that

$$U(F) = \int_Z u(z) dF(z) \quad \text{represents } \succsim.$$

- The function u is called the vNM utility index, or Bernoulli utility function. The function U is the expected utility.
- I will prove this theorem, since the proof is advanced.²

²However, if you are interested in taking this to a greater depth and understanding the gory details, the following references are useful:

- David M. Kreps. *Microeconomic Foundations I: Choice and Competitive Markets*. Princeton University Press, 2013. This is a good place to start.
- David M. Kreps. *Notes on the Theory of Choice*. Westview Press, 1988. If you want to go more in depth.
- Peter C. Fishburn. *Utility Theory for Decision Making*. John Wiley and Sons, 1970. If you want the full details. Difficult to find.

Definition 4.8 (Attitudes towards Risk)

Let $\succsim$ be a preference relation over $\mathfrak{F}$.

- We say that $\succsim$ is **risk averse** if for any lottery p , $[E(p)] \succsim p$, i.e. if the DM prefers the sure lottery that gives as a prize the expected value of the lottery versus receiving the lottery itself.
- We say that $\succsim$ is **risk neutral** if for any lottery p , $[E(p)] \sim p$.
- We say that $\succsim$ is **risk loving** if for any lottery p , $[E(p)] \precsim p$.

Note that if the cdf F represents a lottery and preferences admit an expected utility representation with Bernoulli utility function $u(z)$ a DM is risk averse if and only if

$$\int_{\mathcal{Z}} u(z) dF(z) \leq u\left(\int_{\mathcal{Z}} z dF(z)\right). \quad (4)$$

Risk aversion is equivalent to concavity of u

- Note that the inequality in 4 is a defining property of concave functions, i.e. if preferences are representable a Bernoulli utility function u , then **risk aversion is equivalent to the concavity of u** .
- Recall some basic properties of concave functions:
 - An increasing and concave function must be continuous (but not necessarily differentiable.)
 - Jensen's inequality*. If u is concave, then for any finite collection of numbers $(\alpha_k)_{k=1,\dots,K}$ that add up to 1, $u\left(\sum_{k=1}^K \alpha_k x_k\right) \geq \sum_{k=1}^K \alpha_k u(x_k)$. Note that equation (4) is the “continuous” version of Jensen's inequality, where the weights α have been replaced by the CDF, F . You can see this more easily if you assume that F has a pdf f .
 - Three Strings Lemma*: For any $a < b < c$, we have

$$\frac{u(c) - u(b)}{c - b} \leq \frac{u(c) - u(a)}{c - a} \leq \frac{u(b) - u(a)}{b - a}.$$

- If u is twice differentiable, then for any $a < c$, $u'(a) \geq u'(c)$, and thus $u''(x) \leq 0$ for all x .

Definition 4.9 (Certainty Equivalent and Risk Premium)

Given a Bernoulli utility function, u , we define the following concepts:

- 1 The **certainty equivalent** of a lottery $F \in \mathfrak{F}$, is the amount of money for which an individual is indifferent between F and the certain amount $c(F, u)$; that is, $c(F, u)$ is the number that solves the equation:

$$u(c(F, u)) = \int_Z u(z) dF(z)$$

- 2 The **risk premium** of a lottery $F \in \mathfrak{F}$ is the difference

$$R(F, u) = E[F] - c(F, u),$$

where $E[F]$ denotes the expected value of F , i.e., $E[F] = \int_Z z dF(z)$.

Proposition 4.10

Suppose the decision maker's preferences are represented by an increasing u . The DM is a risk averter if and only if $R(F, u) \geq 0$ for all $F \in \mathfrak{F}$.

Proof.

Let $F \in \mathfrak{F}$. Then we have the following chain of equivalent statements:

$$\begin{aligned} R(F, u) \geq 0 &\Leftrightarrow c(F, u) \leq \int z dF(z) \Leftrightarrow u(c(F, u)) \leq u\left(\int z dF(z)\right) \Leftrightarrow \\ &\int u(z) dF(z) \leq u\left(\int z dF(z)\right) \end{aligned}$$



Application: Insurance

- Consider a strictly risk-averse decision maker who has an initial wealth of w but who runs a risk of a loss of D kzt.
- The probability of the loss is π .
- The DM can buy insurance. One unit costs q kzt and pays 1 kzt if the loss occurs.
- If α units of insurance are bought, the wealth of the individual will be
 - $w - \alpha q$ if there is no loss
 - $w - \alpha q - D + \alpha$ if there is a loss.
- The expected wealth of the DM is then $w - \pi D + \alpha(\pi - q)$.
- The decision maker's problem is to choose the optimal level of α :

$$\max_{\alpha \geq 0} (1 - \pi)u(w - \alpha q) + \pi u(w - \alpha q - D + \alpha)$$

- If α^* is an optimum, it must satisfy the first-order condition:

$$-q(1 - \pi)u'(w - \alpha^* q) + \pi(1 - q)u'(w - D + \alpha^*(1 - q)) \leq 0$$

with equality if $\alpha^* > 0$.

Actuarially fair insurance

- We are going to consider the case of **actuarially fair insurance**, that is, when the cost of insurance is equal to the probability of loss, $q = \pi$.
- Then the first order condition requires that

$$u'(w - D + \alpha^*(1 - \pi)) - u'(w - \alpha^*\pi) \leq 0$$

with equality if $\alpha^* > 0$.

- Since $u'(w - D) > u'(w)$, we must have $\alpha^* > 0$, and therefore

$$u'(w - D + \alpha^*(1 - \pi)) = u'(w - \alpha^*\pi)$$

- Because u' is strictly decreasing due to risk aversion, this implies

$$w - D + \alpha^*(1 - \pi) = w - \alpha^*\pi$$

which implies that $\alpha^* = D$.

- The lesson is: **if insurance is actuarially fair, the DM insures completely.**
- Alternative way: notice that if $q = \pi$, the DM's expected wealth is $w - \pi D$ for any α . Since setting $\alpha = D$ allows him to reach $w - \pi D$ with certainty, the definition of risk aversion implies that this is the optimal level of α .

Application: Demand for a Risky Asset

- An **asset** is a divisible claim to a financial return in the future.
- Suppose there are two assets, a safe asset with a return of 1 kzt per kzt invested and a risky asset with a random return of z kzt per kzt invested.
- The random return has a distribution $F(z)$ that we assume satisfies $\int z dF(z) > 1$, so that its mean return exceeds the one of the safe asset.
- An individual has an initial wealth w to invest.
- Let α and β denote the amounts of wealth invested in the risky and the safe asset, respectively. Thus, for any realization z of the random return, the individual's portfolio (α, β) pays $\alpha z + \beta$.
- The budget restriction is $\alpha + \beta = w$.
- The individual must solve

$$\max_{\alpha, \beta \geq 0} \int u(\alpha z + \beta) dF(z)$$

such that $\alpha + \beta = w$

- Alternatively, the individual's problem is

$$\max_{0 \leq \alpha \leq w} \int u(w + \alpha(z - 1)) dF(z)$$

- If α^* is optimal, it must satisfy the first-order condition:

$$\phi(\alpha^*) = \int u'(w + \alpha^*(z - 1))(z - 1) dF(z) = \begin{cases} \leq 0 & \text{if } \alpha^* < w, \\ \geq 0 & \text{if } \alpha^* > 0. \end{cases}$$

- Note that $\int z dF(z) > 1$ implies that $\phi(0) > 0$. Hence, $\alpha^* = 0$ cannot satisfy this first-order condition.
- Lesson: if a risk is actuarially favorable, then a risk averter will always accept at least a small amount of it.

Comparisons across individuals: the “More risk averse” relation & the coefficient of absolute risk aversion.

- We wish to formalize the statement “one decision maker is more risk averse than another”. Let’s first introduce the following definition.

Definition 4.11

Given a (twice-differentiable) Bernoulli utility function u for money, the Arrow-Pratt coefficient of absolute risk aversion at x is defined as

$$r_A(x, u) = -\frac{u''(x)}{u'(x)}$$

- It turns out that r_A is a measure of risk aversion that is related to the curvature of u .
- We will go back to this point in a bit, but first I will state a proposition.

Proposition 4.12

Consider two individuals utility functions u_1 and u_2 . Then, the following definitions of the *more-risk-averse-than* relation are equivalent: we say that u_1 is more risk averse than u_2 if:

- ① $r_A(x, u_1) \geq r_A(x, u_2)$ for every $x \in Z$.
- ② There exist an increasing concave function Ψ such that $u_1(x) = \Psi(u_2(x))$ at all x ; i.e., u_1 is a concave transformation of u_2 .
- ③ $c(F, u_1) \leq c(F, u_2)$ for any $F \in \mathfrak{F}$.
- ④ Whenever u_1 finds a lottery F at least as good as a risk-less lottery $[c]$, then u_2 also finds F at least as good as $[c]$. That is $\int u_1(z)dF(z) \geq u_1(c)$ implies $\int u_2(z)dF(z) \geq u_2(c)$ for any $F \in \mathfrak{F}$ and $c \in Z$.

In the homework, most likely I will ask you to prove the equivalence between (2), (3) and (4). In here I will prove that (1) and (2) are equivalent.

Proof that (1) and (2) are equivalent.

Proof.

Start with $u_1(x) = \Psi(u_2(x))$. Differentiating we get

$$u_1'(x) = \Psi'(u_2(x))u_2'(x)$$

and

$$u_1''(x) = u_2''(x)\Psi'(u_2(x)) + \Psi''(u_2(x))(u_2'(x))^2$$

Dividing by $-u_1'(x)$ and using the first expression, yields

$$r_A(x, u_1) = r_A(x, u_2) - \frac{\Psi''(u_2(x))}{\Psi'(u_2(x))} u_2'(x).$$

Thus $r_A(x, u_1) \geq r_A(x, u_2)$ if and only if $\Psi''(u_2) \leq 0$ for all u_2 in the range of $u_2 : Z \rightarrow \mathbb{R}$. □

Application: demand for a risky asset, continued.

- We analyze here the same asset portfolio problem as before.
- Suppose that we now have two individuals with Bernoulli utility functions u_1 and u_2 . Denote their optimal choices by α_1^* and α_2^* .
- We will show here that if u_1 is more risk-averse than u_2 then $\alpha_1^* < \alpha_2^*$; i.e., the first decision maker invests less in the risky asset than the second one.
- Repeating from before, the problem of each decision maker $i = 1, 2$ is

$$\max_{0 \leq \alpha \leq w} \int u_i(w + \alpha(z - 1)) dF(Z).$$

- Assuming an interior solution, we have that the optimal solutions for each $i = 1, 2$ satisfy:

$$\phi_i(\alpha_i^*) = \int (z - 1) u_i'(w + \alpha_i^*(z - 1)) dF(z) = 0.$$

Application: demand for a risky asset, continued.

- As we know, the concavity of u_1 implies that ϕ_1 is decreasing. Therefore, if we show that $\phi_1(\alpha_2^*) < 0$, it must follow that $\alpha_2^* > \alpha_1^*$.
- Since $u_1(x) = \Psi(u_2(x))$, we can write

$$\phi_1(\alpha_2^*) = \int (z-1)\Psi'(u_2(w + \alpha_2^*(z-1)))u_2'(w + \alpha_2^*(z-1))dF(z) < 0.$$

- To understand the final inequality, note that the integrand in the expression is the same as in

$$\phi_2(\alpha_2^*) = \int (z-1)u_2'(w + \alpha_2^*(z-1))dF(z) = 0,$$

except that it is multiplied by Ψ' , which is a positive and decreasing function of z . (Recall that, by definition, Ψ is increasing and concave, thus Ψ' is positive and decreasing).

- Hence, the integral underweights the positive values of $u_2'(w + \alpha_2^*(z-1))$, which obtain for $z > 1$, relative to the negative values, which obtain for $z < 1$. Since in the last expression, the integral of the positive and the negative parts of the integrand added to zero, they must now add to a negative number in the former expression.

Comparison of Payoff distributions in terms of return and risk.

- In the previous part of the lecture we focused on comparing utility functions.
- The aim now is to compare payoff distribution.
- In particular, we will try to answer questions of the type: “Given a set of preference relations on $\mathfrak{F}$, for what $F, G \in \mathfrak{F}$ is $F \succsim G$ for all preference relations in the set?”
- We will try to provide an answer to this question by formalizing some orders on the set $\mathfrak{F}$.

First Order Stochastic Dominance

- We want to formalize the expression: “The distribution F yields unambiguously higher returns than the distribution G ”.
- There are two possibilities:
 - We could test whether every expected utility maximizer who values more over less money, prefers F to G .
 - Alternatively, we could verify if for every amount of money x , the probability of getting at least x is higher under F than G .
- Turns out that both ways of looking at this issue are the same.

Definition 4.13

The distribution $F \in \mathfrak{F}$ **first-order stochastically dominates** G , denoted $F \triangleright_{\text{fosd}} G$, if, for every non-decreasing function $u : Z \rightarrow \mathbb{R}$, we have

$$\int u(z) dF(z) \geq \int u(z) dG(z)$$

Proposition 4.14

Let $F, G \in \mathfrak{F}$. Then $F \triangleright_{\text{fosd}} G$ if and only if $F(x) \leq G(x)$ for every x .

Equivalently, define for any cdf F , $\bar{F}(z) = 1 - F(z)$ for every $z \in Z$. Intuitively, $\bar{F}(z) = \Pr(\text{the monetary prize is greater than } z)$. Then. $F \triangleright_{\text{fosd}} G \Leftrightarrow$ if $\bar{F}(x) \geq \bar{G}(x)$ for all x .

Proof about equivalence of FOSD definitions

Proof.

Let $F, G \in \mathfrak{F}$. To simplify the proof, we will assume that the distributions F and G have pdf's f and g respectively. Note that it suffices to establish the result for differentiable utility functions u . Recall that we assumed $Z = [w, b] \subset \mathbb{R}$, where $w < b$. We establish an equivalent form for $E[u(F)]$ first:

$$\begin{aligned}\int_w^b u(z) dF(z) &= \int_w^b u(z) f(z) dz = \int_w^b \left(\int_w^z u'(x) dx + u(w) \right) f(z) dz \\ &= \int_w^b \left(\int_w^z u'(x) dx \right) f(z) dz + u(w) = \int_w^b \int_x^b u'(x) f(z) dz dx + u(w) \\ &= \int_w^b u'(x) \left(\int_x^b f(z) dz \right) dx + u(w) = \int_w^b u'(x) \bar{F}(x) dx + u(w)\end{aligned}$$



Proof.

We prove the $\Leftarrow$ (if) part. Note that:

$$\int_w^b u(z) dF(z) \geq \int_w^b u(z) dG(z) \Leftrightarrow \int_w^b u'(x) \bar{F}(x) dx \geq \int_w^b u'(x) \bar{G}(x) dx.$$

Thus, if $\bar{F}(x) \geq \bar{G}(x)$ and since u is increasing (and thus $u' > 0$), then $\int_w^b u'(x) \bar{F}(x) dx \geq \int_w^b u'(x) \bar{G}(x) dx$ and the if part of the proposition is true.

We now prove the $\Rightarrow$ (only if) part. We use the contrapositive here. Suppose that there exists a prize $\hat{z}$ for which $\bar{F}(\hat{z}) < \bar{G}(\hat{z})$. Define the following utility function $\hat{u}(x) = 1$ for $x > \hat{z}$ and $\hat{u}(x) = 0$ for $x \leq \hat{z}$. This u is non-decreasing. But then $\int_w^b \hat{u}(x) dF(x) = \bar{F}(\hat{z}) < \bar{G}(\hat{z}) = \int_w^b \hat{u}(x) dG(x)$. Thus, by the contrapositive, the assertion is proved. □

Example 4.15

Consider a compound lottery that has as its first stage a realization of x according to G and in its second stage applies to the outcome x of the first stage an *upwards probabilistic shift*. That is, if outcome x is realized in the first stage, then the second stage pays a final amount of money $x + z$, where z is distributed according to $H_x(z)$ with $H_x(0) = 0$. Thus H_x generates a final return of at least x with probability one.

Denote the resulting distribution by F . Then for any nondecreasing $u : Z \rightarrow \mathbb{R}$ we have

$$\int u(t) dF(t) = \int \left(\int u(x + z) dH_x(z) \right) dG(x) \geq \int u(t) dG(t).$$

Thus, $F \triangleright_{\text{fosd}} G$.

- We want now to introduce a comparison based on relative riskiness or dispersion.

Definition 4.16

For any two distributions F and G with the same mean, F **second-order stochastically dominates** G , denoted $F \triangleright_{\text{sosd}} G$, if for every non-decreasing concave function $u : Z \rightarrow \mathbb{R}$, we have

$$\int u(z) dF(z) \geq \int u(z) dG(z)$$

Definition 4.17

For any $F, G \in \mathfrak{F}$, G is a mean-preserving spread of F if and only if

$$Y = X + \epsilon$$

for some random variables $X \sim F$, $Y \sim G$ and ϵ such that $E[\epsilon|X = x] = 0$ for all x .

- Imagine that for every realization of x , we add a noise ϵ and give the decision maker $y = x + \epsilon$. In other words, we are spreading the probabilities without changing the mean.
- If the decision maker is risk averse, he would not like to have this scheme. He would rather have x .
- It is instructive to compare this to example 4.23. In that example, we were giving the decision maker an extra amount of money for every realization x .
- While this could increase the variance of money, the decision maker knew that he was getting more.
- In the current scheme, we are increasing the variance without increasing the expectation.

Proposition 4.18

Assume that $F, G \in \mathfrak{F}$ have the same mean. Then, the following are equivalent.

- ① For every non-decreasing concave function $u : Z \rightarrow \mathbb{R}$, we have $\int u(z) dF(z) \geq \int u(z) dG(z)$
- ② G is a mean-preserving spread of F .
- ③ For every $t \geq 0$, $\int_w^t G(x) dx \geq \int_w^t F(x) dx$.

Proof.

Homework! You can use a similar technique as in the proof of 4.22, by using integration by parts twice. □

Monotone Likelihood ratio property

- Let $F, G \in \mathfrak{F}$ be distributions with densities f and g respectively.
- Define the **odds-ratio**

$$l(x) = \frac{f(x)}{g(x)}.$$

- Suppose that you observe a draw x from a random variable that may have distribution F or G .
- You ask the question: *given the realization x , is it more likely that it came from F than from G ?*
- It turns out that $l(x)$ is the ratio to which you need to multiply the prior odds to get the posterior odds!

Definition 4.19

The pair (f, g) has the MLRP (monotone likelihood ratio property) if $I(x)$ is non-decreasing.

- Intuitively, the higher the realized value of x , the more likely that it was drawn from the high distribution F .
- MLRP implies FOSD, but it is a stronger condition.
- You could have FOSD and still there might be some high signal values that likely come from G .

Example 4.20

Suppose $f(0) = f(2) = 0.5$ and $g(1) = g(3) = 0.5$. Then $G \succ_{\text{fosd}} F$ but the MLRP property fails.

- The MLRP is often used as an assumption in models of moral hazard and screening.
- Let $F(x|a)$ be a family of distribution functions parameterized/indexed by the parameter a .
- The parameter a could be an action (e.g. effort), or a type (e.g. ability) of an agent, and x is the outcome (e.g. the amount produced/consumed).
- The MLRP tells us that if $x_2 > x_1$ and $a_2 > a_1$ then

$$\frac{f(x_2|a_2)}{f(x_2|a_1)} \geq \frac{f(x_1|a_2)}{f(x_1|a_1)}.$$

- In other words, if a principal observes a higher x , it will guess a higher likelihood that it came about due to a higher a .
- This is important in the design of incentives!

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$$\int u(z) dF(z) \geq \int u(z) dG(z)$$

Proposition 4.22

Let $F, G \in \mathfrak{F}$. Then $F \triangleright_{\text{fosd}} G$ if and only if $F(x) \leq G(x)$ for every x .

Equivalently, define for any cdf F , $\bar{F}(z) = 1 - F(z)$ for every $z \in Z$. Intuitively, $\bar{F}(z) = \Pr(\text{the monetary prize is greater than } z)$. Then, $F \triangleright_{\text{fosd}} G \Leftrightarrow$ if $\bar{F}(x) \geq \bar{G}(x)$ for all x .

Proof about equivalence of FOSD definitions

Proof.

Let $F, G \in \mathfrak{F}$. To simplify the proof, we will assume that the distributions F and G have pdf's f and g respectively. Note that it suffices to establish the result for differentiable utility functions u . Recall that we assumed $Z = [w, b] \subset \mathbb{R}$, where $w < b$. We establish an equivalent form for $E[u(F)]$ first:

$$\begin{aligned}\int_w^b u(z) dF(z) &= \int_w^b u(z) f(z) dz = \int_w^b \left(\int_w^z u'(x) dx + u(w) \right) f(z) dz \\ &= \int_w^b \left(\int_w^z u'(x) dx \right) f(z) dz + u(w) = \int_w^b \int_x^b u'(x) f(z) dz dx + u(w) \\ &= \int_w^b u'(x) \left(\int_x^b f(z) dz \right) dx + u(w) = \int_w^b u'(x) \bar{F}(x) dx + u(w)\end{aligned}$$



Proof.

We prove the $\Leftarrow$ (if) part. Note that:

$$\int_w^b u(z) dF(z) \geq \int_w^b u(z) dG(z) \Leftrightarrow \int_w^b u'(x) \bar{F}(x) dx \geq \int_w^b u'(x) \bar{G}(x) dx.$$

Thus, if $\bar{F}(x) \geq \bar{G}(x)$ and since u is increasing (and thus $u' > 0$), then $\int_w^b u'(x) \bar{F}(x) dx \geq \int_w^b u'(x) \bar{G}(x) dx$ and the if part of the proposition is true.

We now prove the $\Rightarrow$ (only if) part. We use the contrapositive here. Suppose that there exists a prize $\hat{z}$ for which $\bar{F}(\hat{z}) < \bar{G}(\hat{z})$. Define the following utility function $\hat{u}(x) = 1$ for $x > \hat{z}$ and $\hat{u}(x) = 0$ for $x \leq \hat{z}$. This u is non-decreasing. But then $\int_w^b \hat{u}(x) dF(x) = \bar{F}(\hat{z}) < \bar{G}(\hat{z}) = \int_w^b \hat{u}(x) dG(x)$. Thus, by the contrapositive, the assertion is proved. □

Example 4.23

Consider a compound lottery that has as its first stage a realization of x according to G and in its second stage applies to the outcome x of the first stage an *upwards probabilistic shift*. That is, if outcome x is realized in the first stage, then the second stage pays a final amount of money $x + z$, where z is distributed according to $H_x(z)$ with $H_x(0) = 0$. Thus H_x generates a final return of at least x with probability one.

Denote the resulting distribution by F . Then for any nondecreasing $u : Z \rightarrow \mathbb{R}$ we have

$$\int u(t) dF(t) = \int \left(\int u(x + z) dH_x(z) \right) dG(x) \geq \int u(t) dG(t).$$

Thus, $F \triangleright_{\text{fosd}} G$.

- We want now to introduce a comparison based on relative riskiness or dispersion.

Definition 4.24

For any two distributions F and G with the same mean, F **second-order stochastically dominates** G , denoted $F \triangleright_{\text{sosd}} G$, if for every non-decreasing concave function $u : Z \rightarrow \mathbb{R}$, we have

$$\int u(z) dF(z) \geq \int u(z) dG(z)$$

Definition 4.25

For any $F, G \in \mathfrak{F}$, G is a mean-preserving spread of F if and only if

$$Y = X + \epsilon$$

for some random variables $X \sim F$, $Y \sim G$ and ϵ such that $E[\epsilon|X = x] = 0$ for all x .

- Imagine that for every realization of x , we add a noise ϵ and give the decision maker $y = x + \epsilon$. In other words, we are spreading the probabilities without changing the mean.
- If the decision maker is risk averse, he would not like to have this scheme. He would rather have x .
- It is instructive to compare this to example 4.23. In that example, we were giving the decision maker an extra amount of money for every realization x .
- While this could increase the variance of money, the decision maker knew that he was getting more.
- In the current scheme, we are increasing the variance without increasing the expectation.

Proposition 4.26

Assume that $F, G \in \mathfrak{F}$ have the same mean. Then, the following are equivalent.

- ① For every non-decreasing concave function $u : Z \rightarrow \mathbb{R}$, we have $\int u(z) dF(z) \geq \int u(z) dG(z)$
- ② G is a mean-preserving spread of F .
- ③ For every $t \geq 0$, $\int_w^t G(x) dx \geq \int_w^t F(x) dx$.

Proof.

Homework! You can use a similar technique as in the proof of 4.22, by using integration by parts twice. □

Monotone Likelihood ratio property

- Let $F, G \in \mathfrak{F}$ be distributions with densities f and g respectively.
- Define the **odds-ratio**

$$l(x) = \frac{f(x)}{g(x)}.$$

- Suppose that you observe a draw x from a random variable that may have distribution F or G .
- You ask the question: *given the realization x , is it more likely that it came from F than from G ?*
- It turns out that $l(x)$ is the ratio to which you need to multiply the prior odds to get the posterior odds!

Definition 4.27

The pair (f, g) has the MLRP (monotone likelihood ratio property) if $I(x)$ is non-decreasing.

- Intuitively, the higher the realized value of x , the more likely that it was drawn from the high distribution F .
- MLRP implies FOSD, but it is a stronger condition.
- You could have FOSD and still there might be some high signal values that likely come from G .

Example 4.28

Suppose $f(0) = f(2) = 0.5$ and $g(1) = g(3) = 0.5$. Then $G \succ_{\text{fosd}} F$ but the MLRP property fails.

- The MLRP is often used as an assumption in models of moral hazard and screening.
- Let $F(x|a)$ be a family of distribution functions parameterized/indexed by the parameter a .
- The parameter a could be an action (e.g. effort), or a type (e.g. ability) of an agent, and x is the outcome (e.g. the amount produced/consumed).
- The MLRP tells us that if $x_2 > x_1$ and $a_2 > a_1$ then

$$\frac{f(x_2|a_2)}{f(x_2|a_1)} \geq \frac{f(x_1|a_2)}{f(x_1|a_1)}.$$

- In other words, if a principal observes a higher x , it will guess a higher likelihood that it came about due to a higher a .
- This is important in the design of incentives!